IBM Developer
SKILLS NETWORK

# Winning Space Race with Data Science

Falcon 9 First-Stage Landing Prediction
FINAL DATA SCIENCE CAPSTONE PROJECT REPORT

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GitHub: https://github.com/AlbyTech-001/testrepo

# IBM Rubric Evidence Map - All 15 Requirements

1-2 | Repository contains 7 completed notebooks, spacex-dash-app.py, CSV data and screenshots: https://github.com/AlbyTech-001/testrepo | Deliverable: this 22-page PDF

3-4 | Executive Summary: 90 launches, 66.7% success, 83.3% model accuracy | Introduction: $62M vs $165M reuse-cost problem

5 | API + Wikipedia + wrangling: normalize JSON/HTML, impute PayloadMass, encode Class and one-hot categories -> 90 labeled records

6-7 | EDA: scatter/bar/line/grouping | Classification: StandardScaler, 80/20 split, 10-fold GridSearchCV, four classifiers

8 | EDA results: later-flight improvement; SSO 5/5, VLEO 12/14, GTO 14/27; annual success reached 90% in 2019

9 | SQL results: 4 site labels; NASA CRS 45,596 kg; F9 v1.1 mean 2,928.4 kg; first ground success 2015-12-22

10-11 | Folium: sites + 90 outcomes + proximity distances | Dash: site dropdown, pie, payload slider and scatter; KSC 10/13

12 | Predictive results: 18 test cases; all four models 83.3%; Tree CV 87.3%; TN=3, FP=3, FN=0, TP=12

13-15 | Conclusion, creativity and insights: false-optimism risk, label harmonization, time-aware validation and geography trade-off

# Executive Summary

- Methods: SpaceX API and web scraping, data wrangling, EDA/SQL, Folium, Plotly Dash and four tuned classifiers.
- 90 Falcon 9 launches were analyzed; 60 first stages landed successfully (66.7%).
- Success improved over time and varied by site and orbit; KSC LC-39A and VAFB SLC-4E were near 77%.
- Logistic Regression, SVM, Decision Tree and KNN each achieved 83.3% test accuracy (15/18).
- The common error profile was TN=3, FP=3, FN=0, TP=12: no success was missed, but false optimism requires risk controls.

# Introduction

Business context: SpaceX advertised Falcon 9 launches near $62M while competing launches could exceed $165M. First-stage reuse is central to that cost advantage.

Problem: predict whether the Falcon 9 first stage will land successfully from public launch data.

Questions answered:

- How do launch site, orbit, payload and program maturity relate to success?
- Can tuned classification models support pricing and launch-risk assessment?
- Which prediction errors create the greatest operational exposure?

# Data Collection Methodology - SpaceX API

GET launches -> normalize JSON -> enrich rocket, payload, launchpad and core IDs -> one row per launch

Fields: FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial and coordinates.

Output: dataset_part_1.csv

GitHub: https://github.com/AlbyTech-001/testrepo
Artifact: jupyter-labs-spacex-data-collection-api.ipynb

# Data Collection Methodology - Web Scraping

Wikipedia HTML -> BeautifulSoup -> locate Falcon 9 tables -> extract cells -> clean dates/payloads -> DataFrame

Captured launch number, date/time, booster, launch site, payload, mass, orbit, customer, mission outcome and landing outcome.

Output: spacex_web_scraped.csv

GitHub: https://github.com/AlbyTech-001/testrepo

Artifact: jupyter-labs-webscraping.ipynb

# Data Wrangling Methodology: 90 Clean, Labeled Records

- Input: Falcon 9 API and web-scraped launch tables.
- Cleaning: filtered Falcon 9, reset FlightNumber, standardized dates/payloads and imputed missing PayloadMass with the mean.
- Target: Class = 1 for successful landing; Class = 0 for failure/no landing.
- Features: one-hot encoded Orbit, LaunchSite, LandingPad and Serial; retained flight, payload, reuse, grid-fin, leg and block variables.
- Outputs: dataset_part_2.csv and model-ready dataset_part_3.csv.
- Validation: 90 labeled records = 60 successes (66.7%) and 30 failures (33.3%).
- Source: https://github.com/AlbyTech-001/testrepo | labs-jupyter-spacex-Data wrangling.ipynb

# EDA Methodology: Charts Tied to Analytical Questions

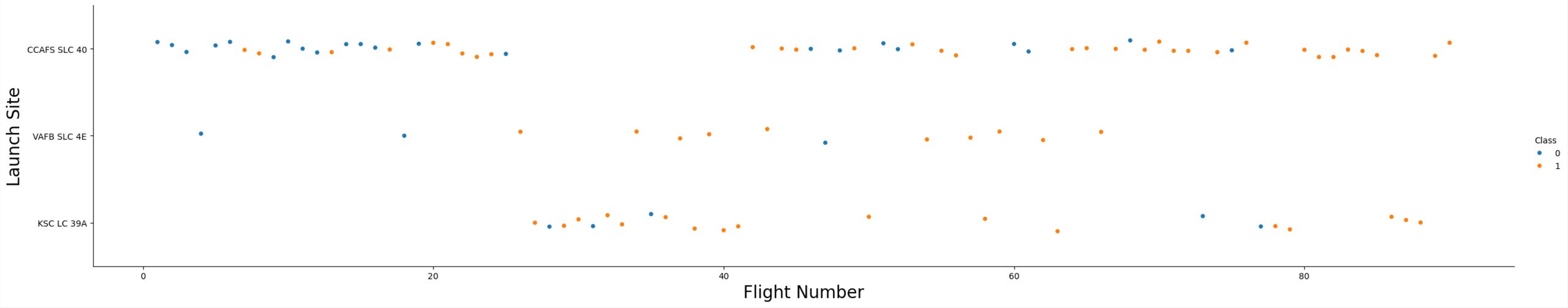
- FlightNumber vs LaunchSite scatter tests whether experience changes success by site.
- PayloadMass vs LaunchSite/Orbit scatter tests whether payload separates outcomes.
- Success-by-Orbit bar chart compares empirical landing rates across mission profiles.
- Annual line chart tests for program learning over time.
- Grouped summaries quantify site, orbit, year and payload bands; sparse groups are not causal proof.
- Source: https://github.com/AlbyTech-001/testrepo | edadataviz.ipynb

# Classification Methodology: Four Models, Tuned Fairly

- Preprocessing: one-hot features standardized with StandardScaler.
- Split: 80% training / 20% test, random_state=2; 18 records held out for final evaluation.
- Models: Logistic Regression, SVM, Decision Tree and K-Nearest Neighbors.
- Tuning: 10-fold GridSearchCV searched model-specific hyperparameters on training folds only.
- Comparison: best CV score, test accuracy and confusion matrix for every model.
- Source: https://github.com/AlbyTech-001/testrepo | SpaceX-Machine-Learning-Prediction-Part-5.ipynb

# EDA Scatter Result: Success Rose with Flight Experience

Method: FlightNumber by LaunchSite, colored by landing Class. Result: later flights show more successes; KSC LC-39A and VAFB SLC-4E were about 77% successful versus 60% at CCAFS SLC-40. Meaning: experience and site add signal. Source: https://github.com/AlbyTech-001/testrepo (edadataviz.ipynb)



# EDA Bar Result: Orbit Is Associated with Success

Method: group Class by Orbit and compute the empirical mean.

Results:

• SSO: 5/5 successes (100%)

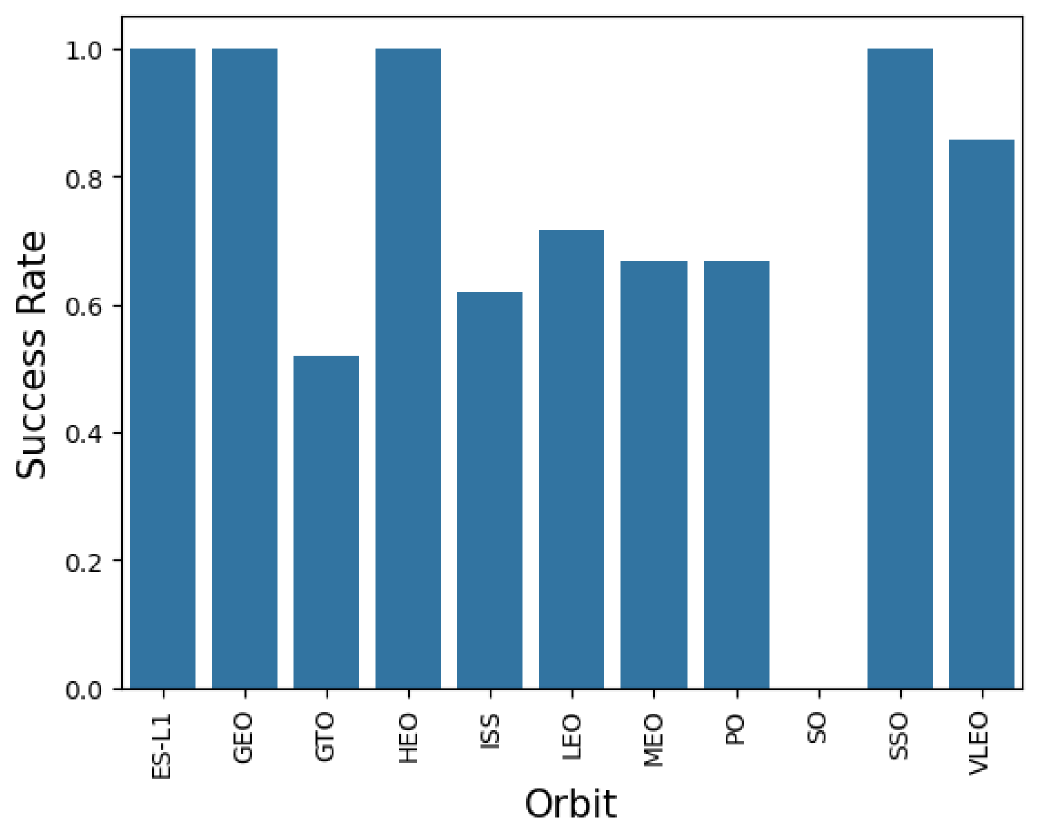
• VLEO: 12/14 (85.7%)

• GTO: 14/27 (51.9%), the largest group

Interpretation: orbit adds context, but perfect rates in one-record groups are not reliable.

GitHub: https://github.com/AlbyTech-001/testrepo

Notebook: edadataviz.ipynb



# EDA Trend Result: Annual Success Rose to 90%

Method: extract launch year, then average binary Class by year.

Results:

• 2010-2013: 0%

• 2017: 83.3%

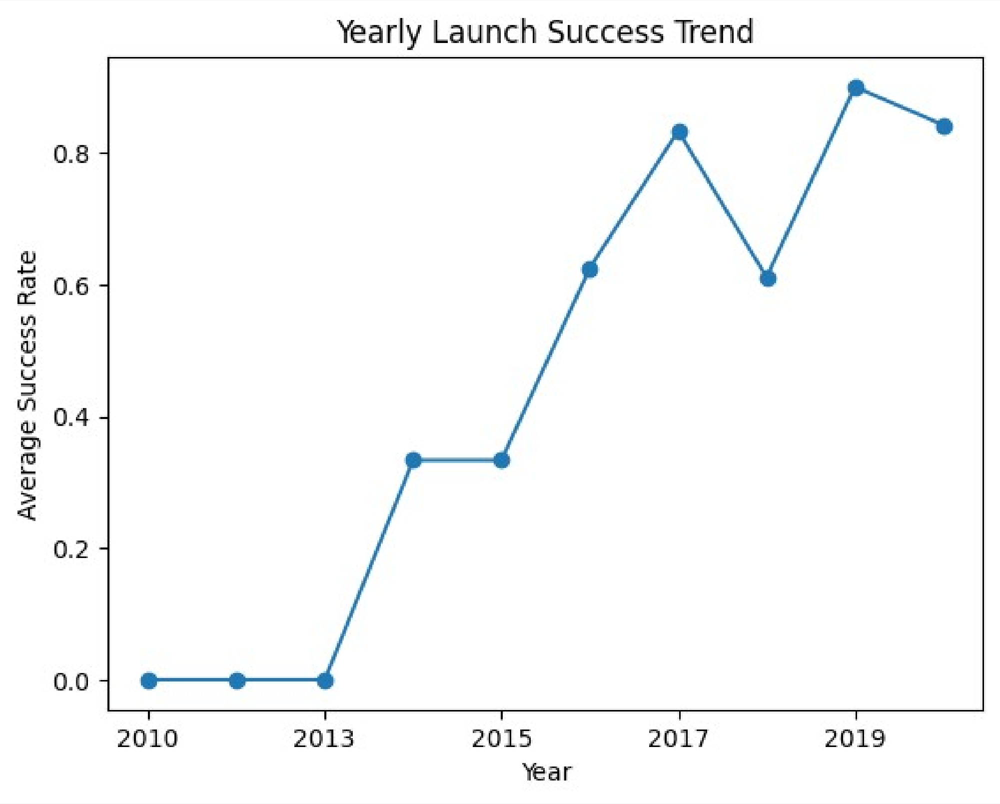
• 2019: 90.0%

• 2020: 84.2%

Interpretation: the upward pattern is consistent with reusable-hardware maturation and operational learning.

GitHub: https://github.com/AlbyTech-001/testrepo

Notebook: edadataviz.ipynb



# EDA with SQL - Executed Queries and Results

SELECT DISTINCT Launch_Site -> CCAFS LC-40, CCAFS SLC-40, KSC LC-39A, VAFB SLC-4E

SELECT SUM(PAYLOAD_MASS__KG_) WHERE Customer LIKE '%NASA (CRS)%' -> 45,596 kg

SELECT AVG(PAYLOAD_MASS__KG_) WHERE Booster_Version='F9 v1.1' -> 2,928.4 kg

SELECT MIN(Date) WHERE Landing_Outcome='Success (ground pad)' -> 2015-12-22

GROUP BY Landing_Outcome -> No attempt 10; drone-ship success 5; drone-ship failure 5

Insight: LC-40/SLC-40 naming variants must be standardized before aggregation.

GitHub: https://github.com/AlbyTech-001/testrepo

Artifact: jupyter-labs-eda-sql-coursera_sqllite.ipynb

# Folium Map Result: All Launch Sites Are Coastal

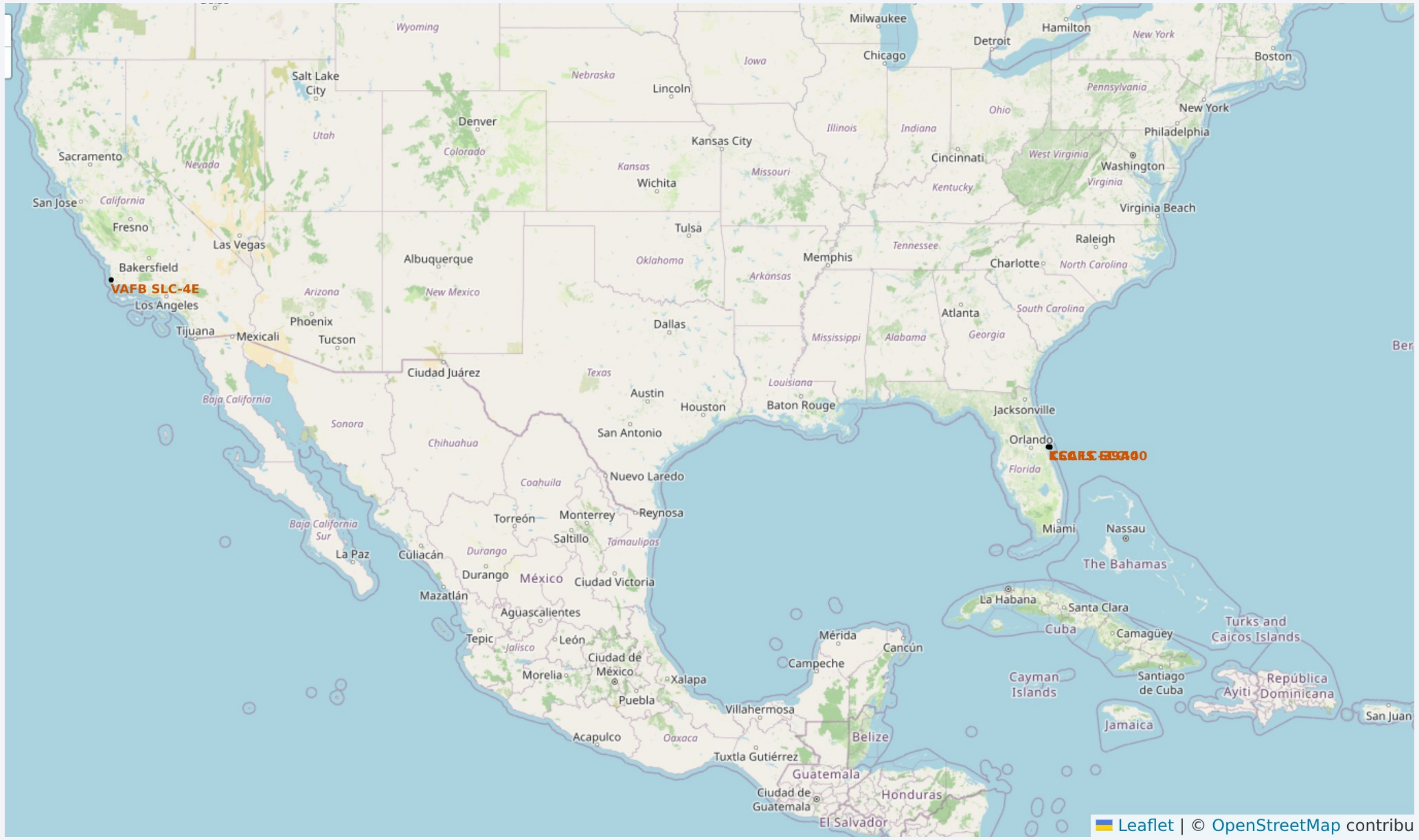
Method: Folium Circle and DivIcon markers plot every launch facility from latitude/longitude.

Result: KSC and CCAFS sites cluster on Florida's Atlantic coast; VAFB is isolated on California's coast.

Interpretation: coastal siting supports downrange safety and ocean recovery logistics.

GitHub: https://github.com/AlbyTech-001/testrepo

Notebook: lab_jupyter_launch_site_location.ipynb



# Folium Records: 90 Launches Show Mixed Outcomes

Method: MarkerCluster expands individual launch records by landing Class.

• Green marker = successful landing

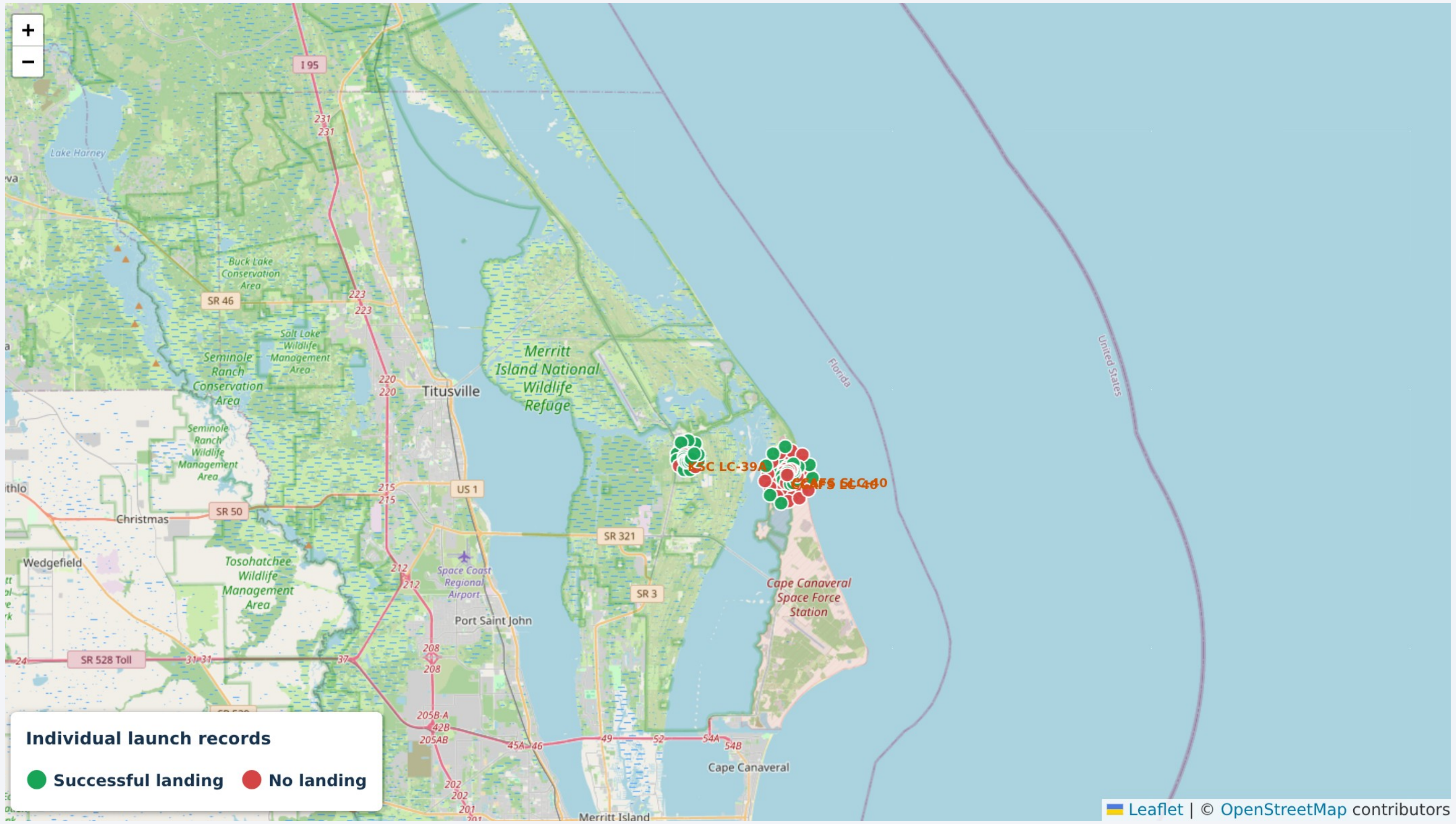
• Red marker = failed/no landing

Result: 90 launch records are mapped - 77 in Florida and 13 at VAFB; KSC and CCAFS contain mixed outcomes.

Interpretation: site alone does not determine success.

GitHub: https://github.com/AlbyTech-001/testrepo

Notebook: lab_jupyter_launch_site_location.ipynb



# Folium Map Result: Access and Safety Are Balanced

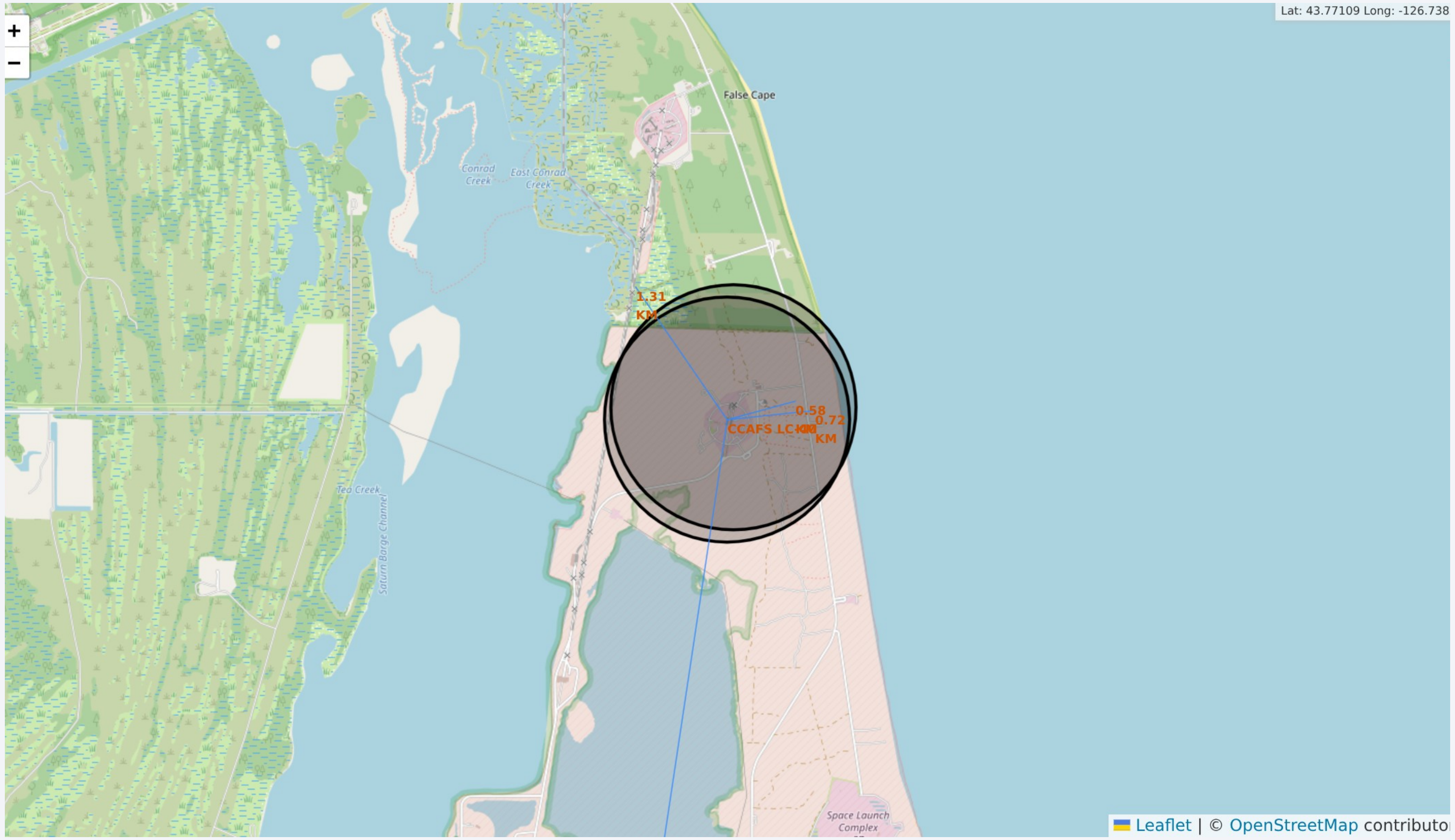
Method: great-circle distance markers and polylines from CCAFS LC-40 to nearby infrastructure.

- Coastline: 0.58 km
- Highway: 0.72 km
- Railway: 1.31 km
- Selected city: 18.14 km

Interpretation: the site combines transport access with separation from dense urban areas.

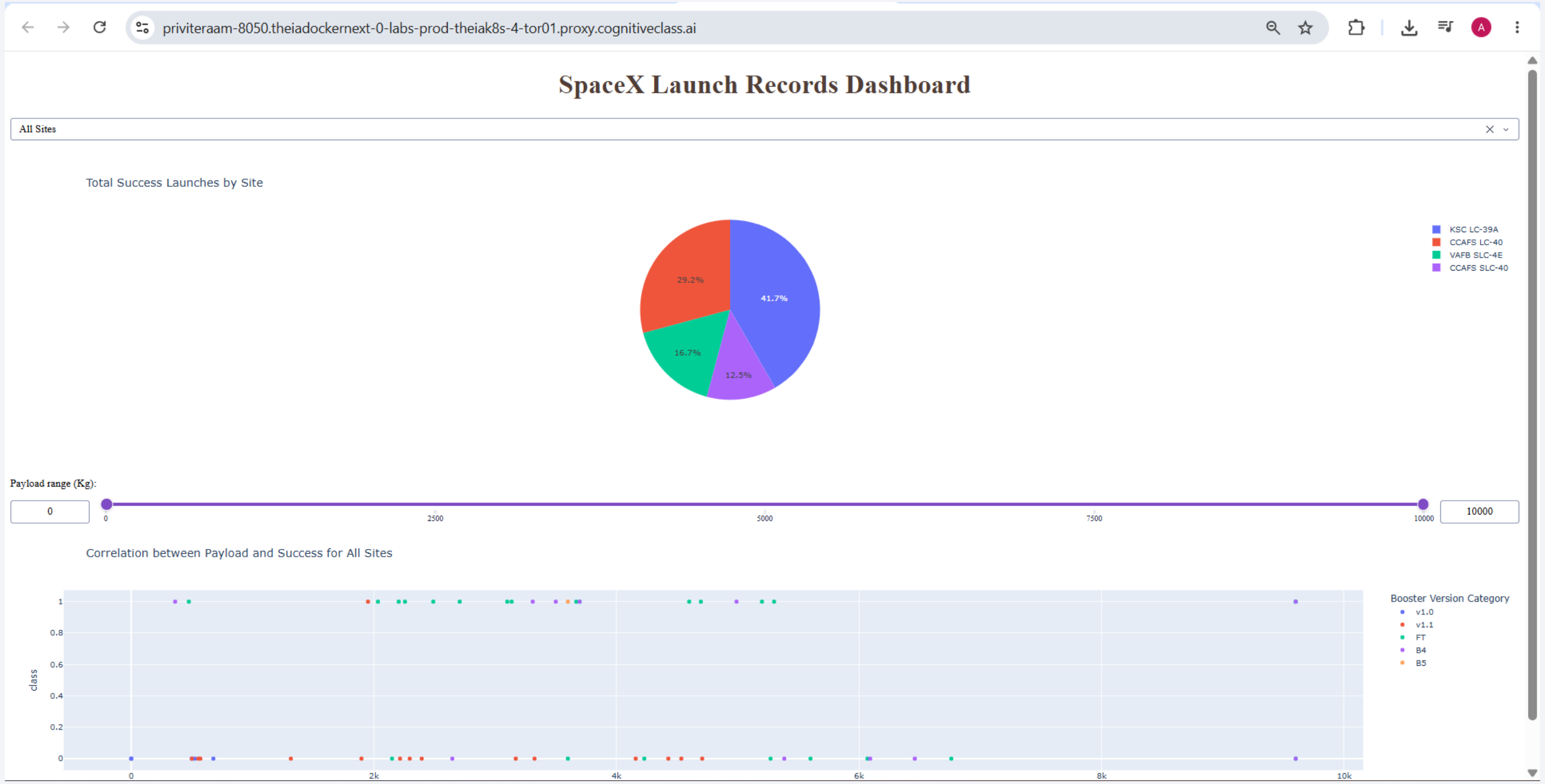
GitHub: https://github.com/AlbyTech-001/testrepo

Notebook: lab_jupyter_launch_site_location.ipynb



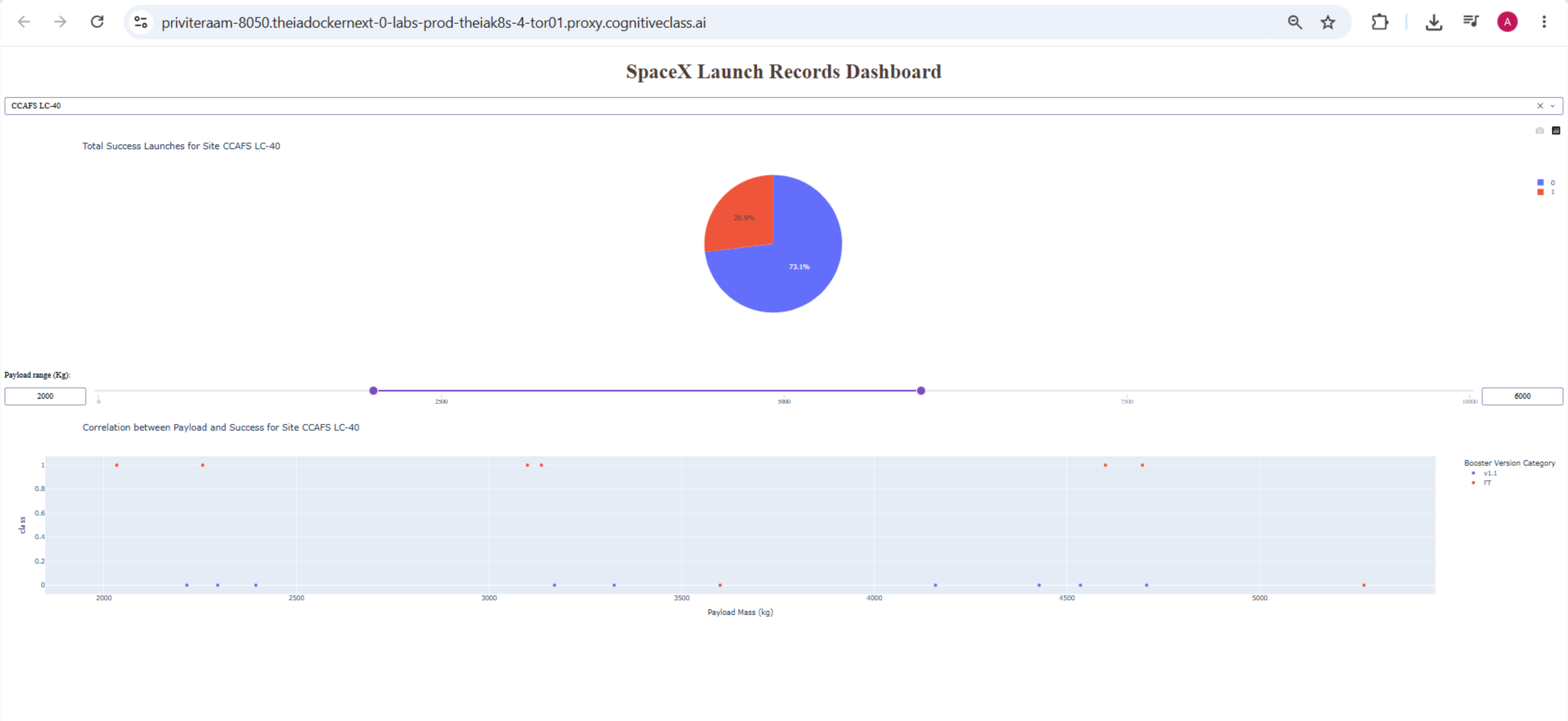
# Plotly Dash Results - All Launch Sites

The interactive dropdown, success pie chart, payload slider and payload-vs-Class scatter are linked by callbacks. The dataset has 24 successes in 56 launches; KSC LC-39A contributed 10 of 24 successes (41.7%). GitHub: https://github.com/AlbyTech-001/testrepo | Artifact: spacex-dash-app.py



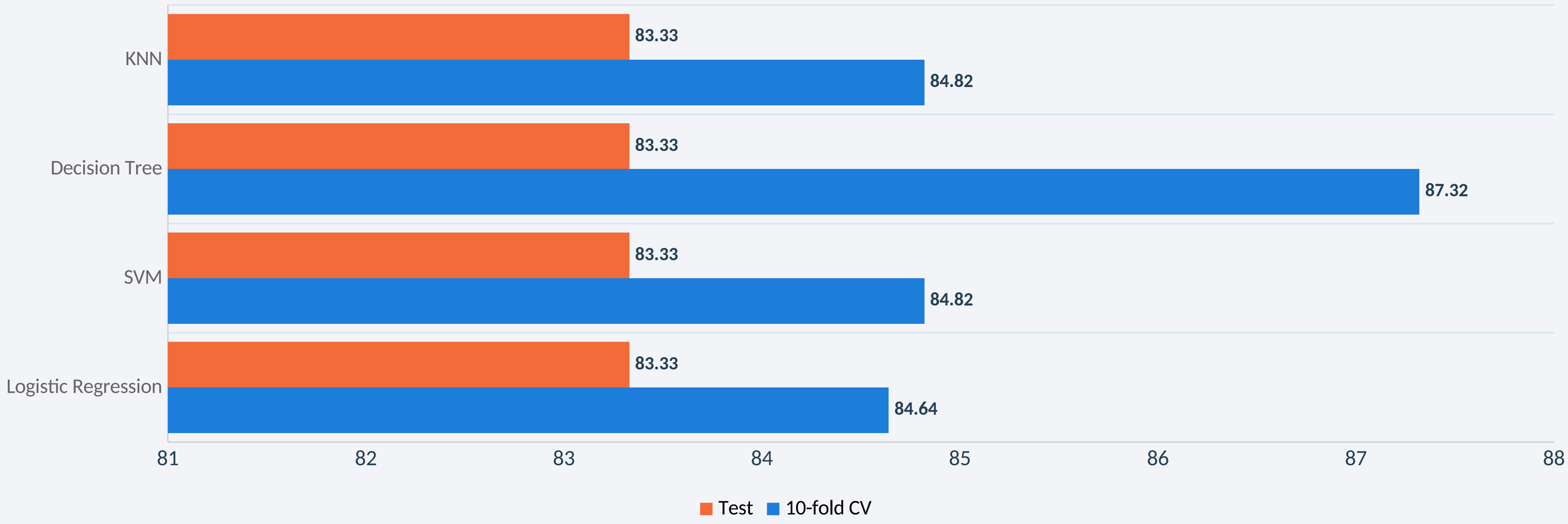
# Plotly Dash Results - Payload Filter

The 2,000-6,000 kg slider filters the scatter while site selection remains active. Mixed outcomes remain visible, showing that payload mass alone is insufficient for prediction.
GitHub: https://github.com/AlbyTech-001/testrepo | Artifact: spacex-dash-app.py



# Predictive Results: All Four Models Scored 83.3%

Same 18-record test set: Logistic Regression, SVM, Decision Tree and KNN each classified 15/18 correctly (83.3%). Decision Tree led 10-fold CV at 87.3%, but no model uniquely won on test. Source: https://github.com/AlbyTech-001/testrepo (SpaceX-Machine-Learning-Prediction-Part-5.ipynb)



# Confusion Matrix: Three False Positives Drive Risk

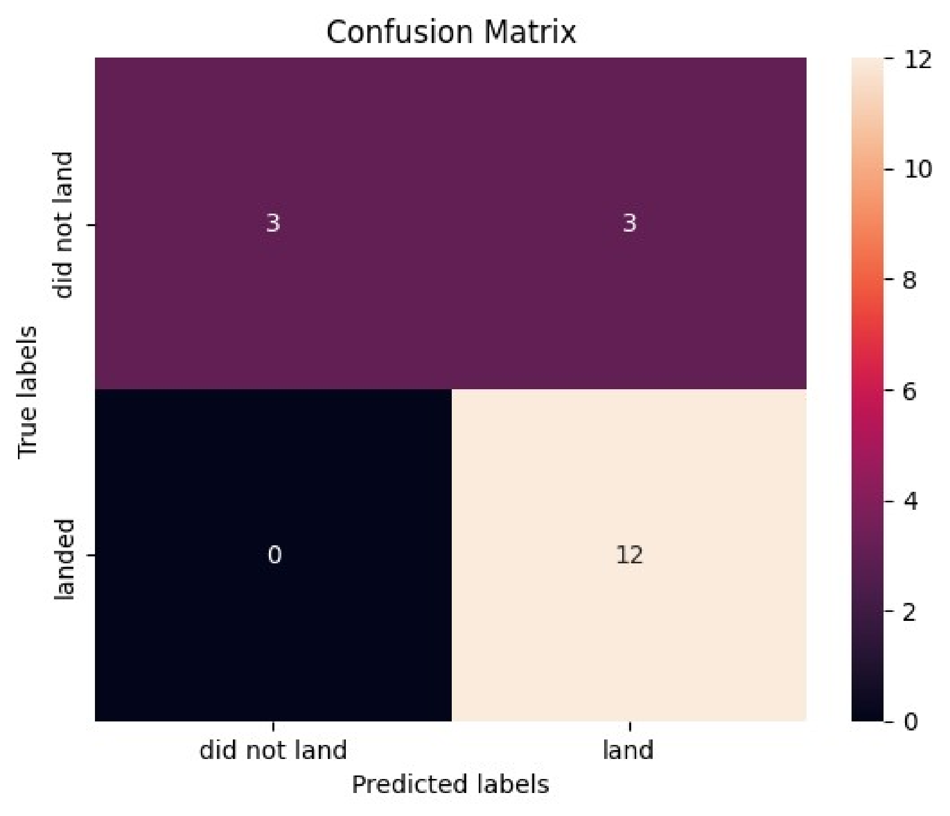
All four models shared this test matrix:

TN = 3 | FP = 3 | FN = 0 | TP = 12

Accuracy = 83.3% (15/18)

Recall = 100% | Precision = 80% | Specificity = 50%

Interpretation: no success was missed; three failures were predicted as successes. This false optimism is the key pricing risk.




Source: https://github.com/AlbyTech-001/testrepo | ML notebook

# Conclusion: Predictive Models Support Reuse Risk

- Landing success improved with flight experience and year, showing strong program-maturity signal.
- Launch site and orbit add useful context; payload mass alone does not reliably separate outcomes.
- KSC LC-39A led the dashboard subset at 10/13; broader EDA placed KSC and VAFB near 77%.
- All four tuned classifiers scored 83.3% on the same 18-record test set, so none is uniquely superior.
- Recommendation: penalize false-success predictions and monitor precision/specificity, not accuracy alone.
- Limitations/next steps: add newer launches, harmonize labels, calibrate probabilities and use time-aware validation.

# Creativity, Innovative Insights and Reproducibility

Repository: https://github.com/AlbyTech-001/testrepo

Contains all completed notebooks, spacex-dash-app.py, CSV data and evidence screenshots.

Final PDF path after upload:
https://github.com/AlbyTech-001/testrepo/blob/main/Data%20Science%20Capstone%20Project%20Report.pdf

CREATIVITY

• A consistent evidence narrative links static EDA, SQL, Folium, Dash and machine-learning results.

INNOVATIVE INSIGHTS

• Error cost: 100% landing recall hides 50% specificity; false optimism is the main pricing risk.

• Data quality: LC-40/SLC-40 and outcome-label variants fragment counts unless standardized.

• Validation: a random split may overstate future performance as hardware evolves; time-aware validation is safer.

• Geography: sub-kilometer coast/highway access plus an 18.14 km city buffer reveals a safety-logistics trade-off.

Limitations: 90 EDA records, 56 Dash records and only 18 test records; observed associations are not causal.